

Active Proprioceptive Recognition of Object Shape and Size with a LEAP Hand

G. Sai Pranav, Harish J. Palanthandalam-Madapusi

Abstract

Most robotic hands rely on cameras or special tactile sensors to recognize objects. This study explores whether the LEAP Hand, a dexterous robotic hand, can recognize objects and estimate their size using its own joint responses, without external vision or dedicated tactile sensors. We developed an AI-based system that enables the LEAP Hand to identify objects by analysing joint positions and movements during grasping, using proprioception. The AI based system is evaluated quantitatively using virtual objects of different shapes and sizes, followed by physical grasping demonstrations using 3D-printed geometric shapes and common household items. For virtual objects, the model achieves 86.7% classification accuracy and a mean absolute size error of 1.02 mm on held-out simulated sizes. The 3D-printed objects include a cube with a side length of 7.5 cm and a sphere with a diameter of 7.5 cm. The household items include a water bottle, a computer mouse, and a small kitchen bowl. The LEAP Hand physically grasps these objects, and the system correctly identifies them in the demonstrated tests. These results demonstrate the potential of proprioceptive sensing for object recognition during robotic grasping without external vision or dedicated tactile sensors.

Keywords: proprioception, active perception, robotic hand, object recognition, size estimation, sensorless perception

1. Introduction

A robot can learn about an object by interacting with it. For a dexterous hand, the positions of its joints while closing around an object depend on the object's shape and size. This makes it possible to use the hand's own movements as a source of information about the object. Such an approach is known as proprioceptive perception. The LEAP Hand is a dexterous, an-

thropomorphic robotic hand developed for robot learning and manipulation (Shaw et al., 2023) and references therein.

Previous studies have shown that the movements and joint positions of a robotic hand can be used to recognize objects. Vásquez and Perdereau (2017) developed proprioceptive shape signatures from joint information and showed that they can be used for shape identification, including estimating object pose and size. Sintov and Meir (2023) proposed a kinesthetic approach to object recognition that does not require dedicated tactile sensors and uses grasp information to recognize objects across different sizes and geometric categories. More recently, Kutsuzawa et al. (2024) investigated learning-based estimation of object shape and stiffness using proprioceptive information from a multi-fingered robotic hand, including joint angles and velocities, without relying on vision, tactile sensors, or force sensors. These studies demonstrate the potential of proprioceptive information for robotic object perception and property estimation.

Building on this work, the present study investigates an AI-based approach using the LEAP Hand. During grasping, the hand closes around an object, and the resulting joint configuration provides information about its shape and size. The proposed system uses the final 16 joint angles to classify object shape and estimate size. The main research question is whether these joint angles can distinguish between different object shapes and estimate their sizes, including sizes not used during training. Keeping the object position and grasp fixed in the simulation allows the relationship between object geometry and the resulting joint configuration to be studied under controlled conditions.

The approach is first evaluated quantitatively using virtual objects of different shapes and sizes. Additional demonstrations are then conducted using the physical LEAP Hand to grasp 3D-printed geometric shapes and common household items. During these demonstrations, the hand’s joint responses are used for object identification. The physical-hand implementation uses motor current as an indicator of contact to stop finger movement. These demonstrations explore whether proprioceptive object recognition can also be applied to physical objects.

The results include quantitative classification and size-estimation performance on held-out simulated objects, together with object-identification results from the physical grasping demonstrations. The simulation metrics and physical demonstrations are considered separately. Together, these investigations examine the potential of using proprioceptive information to

recognise objects during robotic grasping without relying on external vision or dedicated tactile sensors.

2. Experimental Setup and Methodology

We used a four-fingered, 16-actuator LEAP Hand represented by its Unified Robot Description Format (URDF) model and link meshes in MuJoCo (Todorov et al., 2012). The hand design follows (Shaw et al., 2023). The joint vector was ordered as index joints 0–3, middle joints 4–7, ring joints 8–11, and thumb joints 12–15. We expressed commanded and recorded angles in degrees, although MuJoCo stores joint coordinates in radians. We expanded the URDF limits at joints 0, 8, 13, and 14 to accommodate the configured grasp. No joint-angle offset was applied between the simulated and physical hands.

We used the same standardised grasp for all simulated objects. Each finger moved from its initial configuration towards a prescribed target in one-degree increments. Contact at any link stopped all four joints of that finger, while the other fingers continued moving. The grasp ended when all fingers had stopped or after 250 iterations.

2.1. Simulation and dataset

We imported cube, sphere, and cylinder models from STEP files using CadQuery and converted them into triangle meshes for MuJoCo. The objects were held fixed at the same position and orientation. We specified explicit object–hand collision pairs and associated contact geometries with their corresponding fingers. Initial object–hand contact and contact with the palm or an unexpected stationary finger were treated as invalid scenes.

We advanced one active finger at a time and checked each proposed posture for collision. When contact occurred, eight bisection steps were used to locate a nearby collision-free final angle. The simulator directly set joint positions; it did not model motor dynamics, contact forces, motor currents, or a learned grasp policy.

For each simulated grasp, we recorded the final 16-joint angle vector,

$$\mathbf{q} = (q_0, q_1, \dots, q_{15}) \in \mathbb{R}^{16},$$

along with the object class, scale factor, reference dimension, and stopping reason. We also saved the grasp trace for replay.

We used 21 scale factors from 0.50 to 1.50 in increments of 0.05 for each of the three object classes, producing 63 scenes. Each class-size combination had one deterministic grasp. Object-pose randomisation and repeated grasps were not included.

2.2. Neural network and training

The model used only the final 16 joint angles as inputs. Joint velocities, motor currents, fingertip trajectories, and contact histories were not used as features. Each angle was normalised using its configured start and target values:

$$x_j = \frac{q_j - a_j}{b_j - a_j},$$

where a_j and b_j are the start and target angles. When these values were equal, the denominator was set to one.

The fully connected neural network had 16 input units, one 32-unit hidden layer with a hyperbolic tangent activation, and four outputs. Three outputs represented object classes, while the fourth predicted the continuous scale factor. A softmax operation converted the class logits into probabilities. The network contained 676 trainable parameters.

We minimised the sum of mean class cross-entropy loss and mean squared scale-prediction error, with a scale-loss weight of one. Training used full-batch Adam for 1000 epochs, a learning rate of 0.02, and random seed 7.

To represent joint-position variation, we generated 50 noisy copies of each simulated sample by adding independent, zero-mean Gaussian noise with a standard deviation of 0.5° to each joint angle. We retained the original sample. This augmentation did not represent systematic calibration errors.

2.3. Validation and error estimation

We split the dataset by object size before noise augmentation. For each class, the scale factors 0.50, 0.75, 1.00, 1.25, and 1.50 were held out for validation. The remaining 16 sizes per class gave 48 training scenes and 15 held-out scenes across the three classes.

The augmented training set contained 2448 samples, while validation used the 15 original held-out samples. We subsequently trained a separate model on all 63 scenes and their noisy copies, giving 3213 samples, for use with the physical hand. All reported quantitative validation metrics refer to the model trained on the 48 training scenes.

Classification accuracy was calculated as the fraction of held-out scenes assigned to the correct class. Size error was evaluated using the mean absolute error in physical units:

$$\text{MAE}_{\text{size}} = \frac{1}{15} \sum_{i=1}^{15} |\hat{s}_i - s_i| d_{c_i},$$

where $\hat{s}_i$ and s_i are the predicted and true scale factors, respectively, and d_{c_i} is the reference dimension of the true class. This metric evaluates the predicted scale even when the predicted class is incorrect; it is not the error of the final class-conditioned size estimate in such cases.

2.4. Physical objects’ identification

The objects included a 3D-printed cube with a side length of 7.5 cm, a 3D-printed sphere with a diameter of 7.5 cm, a water bottle, a computer mouse, and a small kitchen bowl. The system correctly identified these objects in the demonstrated tests.

We tested the system using a 3D-printed cube with a side length of 7.5 cm, a 3D-printed sphere with a diameter of 7.5 cm, a water bottle, a computer mouse, and a small kitchen bowl. The system correctly identified all five objects in the demonstrated tests.

During grasping, motor current was used as a contact indicator to stop finger movement. The measured joint angles were then used for object identification. The implementation and associated codes are available in Ex-ARM (Pranav, 2026).

3. Results and Discussion

We evaluated the proposed method using held-out simulated object sizes and physical grasping demonstrations. The simulation results assess shape classification and size estimation quantitatively, while the physical tests examine object identification using the LEAP Hand.

3.1. Virtual object evaluation

Of the 15 held-out simulated samples, the model correctly classified 13, achieving an accuracy of 86.7%. All five cube samples were classified correctly, while four of the five sphere samples and four of the five cylinder samples were correctly classified. The scale-factor regression achieved a ground-truth-dimension-weighted mean absolute error (MAE) of 1.02 mm. This

metric evaluates the predicted scale factor using the reference dimension of the true object class, even when the predicted class is incorrect.

Table 1 summarises the classification results and the output of the rejection rule. The rule accepted five of the 15 held-out samples, and all five were correctly classified, giving an acceptance rate of 33.3%. The remaining ten samples were rejected. Five were rejected because the angle vectors of the two nearest classes were similar, while the other five were rejected because the maximum class probability was below 0.75.

Although the rejection rule excluded both incorrectly classified samples, it also rejected eight samples that the raw classifier had classified correctly. Thus, the rule reduced the risk of accepting incorrect predictions in this evaluation, but at the cost of rejecting many otherwise correct predictions. Furthermore, the rejection threshold was derived using the same held-out sizes used for evaluation. Its ability to identify unfamiliar objects therefore requires independent testing.

Table 1: Classification and rejection results for the 15 held-out simulated samples. Accepted indicates that the sample passed all four rejection checks.

True class	Scenes	Raw class correct	Accepted (correct)
Cube	5	5	3 (3)
Sphere	5	4	1 (1)
Cylinder	5	4	1 (1)
Total	15	13	5 (5)

The simulation results indicate that the final joint angles contain useful information for distinguishing geometric classes and estimating object scale. However, the classification errors also show that different objects can produce similar final joint configurations. This suggests that recognition based on a single final configuration may be ambiguous when the joint-angle patterns of different classes overlap.

3.2. *Physical object evaluation*

To examine the applicability of the approach beyond simulation, physical grasping demonstrations were conducted using five objects: a 3D-printed cube with a side length of 7.5 cm, a 3D-printed sphere with a diameter of 7.5 cm, a water bottle, a computer mouse, and a small kitchen bowl. The

system correctly identified all five objects in the demonstrated tests. During grasping, motor current was used as an indicator of contact to stop finger movement. The measured joint angles were then supplied to the recognition system. Motor current was used only for contact detection and was not an input feature of the neural network.

These demonstrations extend the investigation to physical objects, including household items beyond the three geometric classes used in the simulation. However, the physical results are qualitative and do not establish a general recognition accuracy. Repeated trials and variations in object pose, placement, and grasp conditions are needed to assess the reliability of the physical implementation.

The main contribution of this study is a compact proprioceptive recognition model that predicts object class and continuous scale from the final 16 joint angles of a LEAP Hand. With 676 trainable parameters, the model uses a relatively small neural network and does not require camera images or dedicated tactile-sensor measurements as inputs. The combination of quantitative evaluation on held-out simulated sizes and physical grasping demonstrations provides a practical framework for investigating object recognition through the joint configuration produced during grasping.

Previous studies have demonstrated the potential of proprioceptive information for object recognition and property estimation. Vásquez and Perdereau (2017) used proprioceptive shape signatures to identify object shape and estimate pose and size. Sintov and Meir (2023) investigated kinaesthetic object recognition without dedicated tactile sensors, while Kutsuzawa et al. (2024) explored learning-based estimation of object shape and stiffness using proprioceptive information. The present study builds on this line of research by using a compact neural network to predict geometric class and continuous scale from the final joint-angle vector alone. Its contribution lies in this specific implementation and its evaluation in simulation and physical grasping demonstrations, rather than in introducing proprioceptive object recognition itself.

Several limitations should be considered. The simulated objects were grasped at a fixed position and orientation, and each class–size combination was represented by a single deterministic grasp. Although joint-angle noise was added during training, this augmentation does not capture the full variability of physical grasps or systematic calibration errors. The quantitative evaluation was also based on only 15 held-out samples, limiting the strength of conclusions about generalisation. In addition, the rejection rule requires

independent validation, and the physical demonstrations were not accompanied by repeated trials or quantitative performance measures. Future work should therefore investigate different object poses and grasp configurations, conduct repeated physical trials, and evaluate the model on additional objects under independent test conditions. Such experiments would help establish the robustness of the approach and its ability to generalise beyond the simulated training classes.

4. Conclusion

This study investigated object recognition and size estimation using the final 16 joint angles of a LEAP Hand, without requiring camera or dedicated tactile sensor inputs. In simulation, the model correctly classified 13 of 15 held-out samples, achieving an accuracy of 86.7%, with a ground-truth-dimension-weighted mean absolute size error of 1.02 mm. The rejection rule accepted five samples, all of which were correctly classified, but rejected most of the held-out samples. Physical demonstrations showed correct identification of five objects, including 3D-printed geometric shapes and common household items.

These results demonstrate the potential of proprioceptive information for object recognition and size estimation during grasping. However, the limited number of simulated tests, fixed grasp conditions, and qualitative nature of the physical demonstrations restrict the conclusions that can be drawn about generalisation and reliability. Further work should include repeated physical trials, varied object poses and sizes, and independent tests on unseen objects. These tests will help establish the robustness of the approach and its ability to identify objects that fall outside the trained classes.

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